THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** tissue	level of organisation
2	**(a)** intercalary meristem	at the nodes/base
3	**(a)** epidermis	single layer
4	**(b)** parenchyma	storage
5	**(c)** sclerenchyma	fibres
6	**(d)** phloem	food conduction
7	**(c)** connective tissue	fluid matrix
8	**(c)** adipose tissue	insulation and energy
9	**(d)** neuron	conducts impulses
10	**(b)** fixed joints	no movement

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A group of similar cells performing a specific function (1); increases efficiency (1).	1+1
12	Tissues of actively dividing cells (1); at the growing tips and along the circumference of stems (1).	1+1
13	Apical (length), lateral (girth), intercalary (regrowth at nodes) (2).	2
14	They are dividing rapidly and need dense cytoplasm, so vacuoles are absent (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Two onion bulbs in jars of water (1); cut the root tips of one jar (1); the cut roots stop elongating, showing apical growth zones (1).	1+1+1
16	Apical: root and shoot tips; lateral: along the stem circumference; intercalary: at the base or nodes (1+1+1).	1+1+1
17	Small, thin walls, large nucleus, dense cytoplasm, no vacuoles, tightly packed (2); suited for rapid division (1).	2+1
	OR: The apical meristem is removed so elongation stops (1.5); lateral buds at the nodes grow into branches (1.5).	1½+1½
18	Single layer of protective cells (1); stomata in leaves for gas exchange and transpiration (1); root hairs for absorption (1).	1+1+1
	OR: Water evaporates through stomata (1); creates a pull in xylem which draws water up (1); removes waste and cools the plant (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Group of similar cells with a function (1). (ii) Cells → tissues → organs → organ systems → organism (2). (iii) Efficiency and specialisation (2).	1+2+2
	OR: (i) Apical, lateral, intercalary with sites and functions (2). (ii) Cutting root tips of an onion stops root elongation (2). (iii) Thin walls, dense cytoplasm, large nucleus, no vacuoles (1).	2+2+1
20	(i) Division of cells in a ring in the stem (1). (ii) Lateral meristem (1). (iii) By counting annual rings (1). (iv) Wide rings mean good growth years; narrow rings dry or cold years (2).	1+1+1+2
	OR: (i) The outermost protective layer (1). (ii) Pores for gas exchange with guard cells controlling opening (2). (iii) Loss of water vapour; transpiration pull, cooling and removal of waste (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** cell → tissue → organ → organ system	hierarchy
2	**(d)** dense cytoplasm and a prominent nucleus	rapid division
3	**(a)** stomata	guard cells
4	**(c)** aerenchyma	buoyancy
5	**(a)** xylem	from roots to leaves
6	**(c)** cork cambium	protective tissue of old stems
7	**(a)** tendons	strong cord-like
8	**(c)** cardiac and involuntary	striated, branched
9	**(c)** ball and socket joint	free movement in all directions
10	**(c)** 12	attached to the sternum

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rings of new wood formed each year by the lateral meristem (1); the age of the tree and climate (1).	1+1
12	The process by which newly formed cells become specialised to perform specific functions (2).	2
13	Protection from injury and water loss; gas exchange through stomata; absorption by root hairs (1+1).	1+1
14	Loss of water vapour through stomata (1); creates transpiration pull and removes wastes (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Parenchyma: living, thin-walled, storage; collenchyma: living, corner thickening, flexibility; sclerenchyma: dead, lignin, strength (1+1+1).	1+1+1
16	Collenchyma gives flexibility to the young stem (1.5); sclerenchyma fibres in coconut husk are rigid and lignified (1.5).	1½+1½
17	Complex tissue of tracheids, vessels, fibres and parenchyma (1.5); tracheids and vessels are dead, tubular and thick-walled (1.5).	1½+1½
	OR: Sieve tubes, companion cells, phloem parenchyma and fibres (1.5); transports food from leaves to other parts (1.5).	1½+1½
18	Roots would starve (1); the phloem is the transporting tissue; the plant would die (2).	1+2
	OR: Xylem vessels are dead tubes (1); transpiration pull from the leaves works together with the water column (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Sclerenchyma (1). (ii) Collenchyma (1). (iii) Sclerenchyma: dead, lignified, rigid; collenchyma: living, flexible (2). (iv) It would be rigid and break rather than bend (1).	1+1+2+1
	OR: (i) Loosely packed living cells; corner-thickened living cells; dead lignified fibres and sclereids (3). (ii) Storage; flexibility; strength (2).	3+2
20	(i) Conducts water vs food; dead vs living; direction upward vs both ways; tracheids/vessels vs sieve tubes (3). (ii) Transpiration pull and cohesion of water in the xylem (2).	3+2
	OR: (i) Transpiration (1). (ii) Xylem (1). (iii) Guard cells at stomata (1). (iv) Stomata close and water is absorbed (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.